



# Chapter 2

# Introduction to HTML5

Internet & World Wide Web  
How to Program, 5/e

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## OBJECTIVES

In this chapter you'll:

- Understand important components of HTML5 documents.
- Use HTML5 to create web pages.
- Add images to web pages.
- Create and use hyperlinks to help users navigate web pages.
- Mark up lists of information.
- Create tables with rows and columns of data.
- Create and use forms to get user input.



# 2.1 Introduction

(超文本)

- ▶ **HTML5 (HyperText Markup Language 5, 超文本標記式語言)**
  - specifies the *structure* and *content* of documents that are displayed in **web** browsers
    - 註：結構分成：標題、內文、章節、段落等
- ▶ We introduce some **basics** HTML5 techniques such as:
  - **tables (表格)**, which are particularly useful for **structuring information (組織性/結構化資訊)** from **databases (如：課表、教學意見、商品陳列)**
  - **forms (表單/收集)** for **submit** information from web users
  - **internal linking (內部連結)** for easier page navigation
  - **meta elements (網頁資訊，如：編碼、關鍵字)** for specifying information about a document



## 2.2 Editing HTML5

- ▶ We can create HTML5 documents by a text editor (such as Notepad, Notepad++, vi, emacs) and saving it with the .html or .htm.

網頁元件：

- Web servers (網站伺服器) : Computers to store (存放) HTML5 documents.
- Clients (客戶端) :
  - E.g. web browsers (on local computer or smartphone) request specific resources of HTML5 documents from web servers.
- ★補充資訊(取自網路)：網頁編輯可以用一般文字編輯器（如：[Visual Studio Code](#)、[Notepad++](#)、[Sublime Text](#)、[Atom](#)、[GNU Emacs](#)、[VIM](#)），也可使用混合式編輯器（如：[Dreamweaver](#) 或 [WebStorm](#)），根據自己的喜好即可。
- 💡參考連結：[網頁設計常用軟體\(2021~\)](#)、[10款HTML5開發工具](#)

## 2.3 First HTML5 Example

- ▶ Example: `ch02/main.html`,
  - displays the message: `Welcome to HTML5!`

The diagram illustrates the structure of a first HTML5 example. It shows a code editor window with the following HTML code:

```
1 <!DOCTYPE html>
2
3 <!-- Fig. 2.1: main.html -->
4 <!-- First HTML5 example. -->
5 <html>
6   <head>
7     <meta charset = "utf-8">
8     <title>Welcome</title>
9   </head>
10
11   <body>
12     <p>Welcome to HTML5!</p>
13   </body>
14 </html>
```

Annotations and explanations:

- 雙標籤 : `<..> </..>`
- //可看得出結構性及內容
- (head)
- (body)
- Tab shows contents of title element
- Content of body

The rendered output in a browser window shows the title "Welcome" and the body content "Welcome to HTML5!".

Fig. 2.1 | First HTML5 example

## 2.3 First HTML5 Example

**<!DOCTYPE html> Document Type Declaration (宣告型態)**

- ▶ The “**DOCTYPE**” is required for browsers to render the page in standards mode.
  - ▶ 註：告知瀏覽器這是一個html

**<!-- ... --> Comments (註解)**

- ▶ To improve readability and describe the content of a HTML5 document.
  - ▶ 註：描述網頁內容，增加程式可讀性
- ▶ Start with **<!--** and end with **-->**.
- ▶ The browser **ignores** the comments when rendering.

```
main.html x
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.1: main.html -->
4  <!-- First HTML5 example. -->
5  <html>
6  <head>
7      <meta charset = "utf-8">
8      <title>Welcome</title>
9  </head>
10
11 <body>
12     <p>Welcome to HTML5!</p>
13 </body>
14 </html>
```



## 2.3 First HTML5 Example

- 雙標籤：`<..> </..>`  
//類似C程式的左右括弧：`{....}`

`<head>...</head>` 標頭

`<body>...</body>` 內文

- ▶ Html **enclose** (圍住) the **head** section and **body** section

- ▶ Thus, **head** and **body** are **nested elements** (可讀性 巢狀/階層狀元件).

- **head section**: contains **document** information, e.g.,
  - **character set** (網頁的編碼類型，如：UTF-8萬國碼，繁體/簡體/英文等)
  - `<title> ..</title>` (網頁“在瀏覽器上”的標題).

Ex: 

```
<head>
  <meta charset = "utf-8">
  <title>Welcome</title>
</head>
```

- ★補充：UTF-8 ( 8-bit Unicode Transformation Format ) 是一種常見文字編碼方式，有興趣的同學可以[點擊](#)進一步了解。

- **body section**: contains the **page's content** (使用者看到的內容)



## 2.3 First HTML5 Example (cont.)

**<>...</> Start Tags and End Tags** (雙標籤)

▶ Each **element** in HTML5 documents uses a “start tag” and “end tag”.

- **start tag**: includes the **element name** in **angle brackets** **<>** (名稱及括號)
  - Ex: **<html>**
- **end tag**: consists of the **element name** with a **forward slash** **(/)** in angle brackets
  - Ex: **</html>**

▶ **Start tags** may have **attributes** (夾帶特殊屬性/參數) to provide additional information.

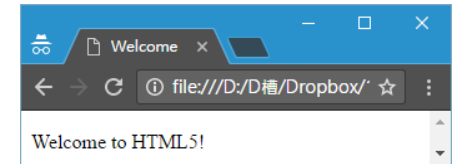
- Ex: **<font size=18 color=blue> Test </font>**
- Each **attribute** has a **name** and a **value** separated by an equals sign (=).
- Ex: **size = 18**
- Ex: **color=blue**
- ▶ “**void elements**” (單標籤，或<.../>): do not have **end tags**.
  - Ex: 換行**<br/>**



## 2.3 First HTML5 Example (Cont.)

### ▶ *<title>...</title>* 網頁標題

- Appear in the **title bar** (標題欄) of the browser
- For the text when **adding** the page to their **Favorites** or **Bookmarks**.
  - 註：作為“我的最愛”或“書籤”的名稱
- Also shown by **Search engines** when displaying results
  - 註：標題也顯示於“搜尋引擎結果”



### ▶ *<p>...</p>* 段落

- All text placed between the `<p>` and `</p>` tags forms **one paragraph**.
  - 註：描述一個段落



```
<body>
  <p>Welcome to HTML5!</p>
</body>
```



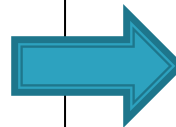
## 2.4 W3C HTML5 Validation Service

- ▶ HTML5 documents that contain **syntax errors** **may not** display properly
  - 註：語法錯誤可能導致呈現錯誤
- ▶ **Validation services** (線上驗證html語法)
  - <http://validator.w3.org>
  - ensure HTML5 documents **syntactically** (語法) correct.
- ★(補充) 網頁Debug小技巧：可使用 Chrome 按[F12] 開啟網頁程式偵錯 (出現"x"表示為錯誤語法所在)

# 2.5 Headings 標題

- ▶ **<h1>..</h1>** heading elements (內文標題)
  - HTML5 provides **six heading elements** (h1, h2, h3, h4, h5, and h6) for *relative importance* of information
    - 註：字體大小從大到小，用於強調重要性

```
1 <!DOCTYPE html>
2
3 <!-- Fig. 2.2: heading.html -->
4 <!-- Heading elements h1 through h6. -->
5 <html>
6   <head>
7     <meta charset = "utf-8">
8     <title>Headings</title>
9   </head>
10
11   <body>
12     <h1>Level 1 Heading</h1>
13     <h2>Level 2 heading</h2>
14     <h3>Level 3 heading</h3>
15     <h4>Level 4 heading</h4>
16     <h5>Level 5 heading</h5>
17     <h6>Level 6 heading</h6>
18   </body>
19 </html>
```



## 2.6 Linking



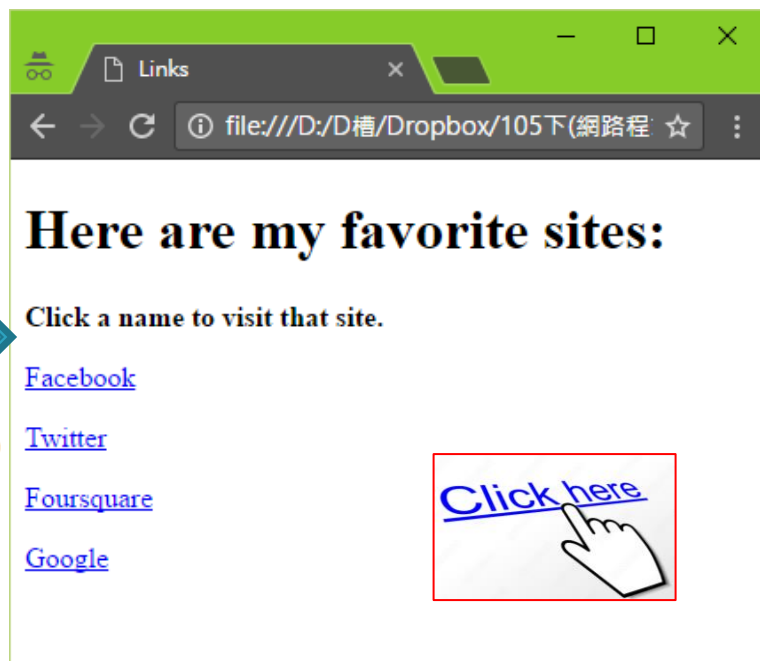
包起來

Linking (網頁連結)

`<a href="連結網址">顯示文字</a>`

- ▶ A **hyperlink** (超連結) references or **links** to other resources, e.g., **HTML5 documents** or **images**.
  - Shown as underline text hyperlinks and color them *blue* by default.
  - 註：超連結通常以藍色底線顯示。

```
3) links.html x
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.3: links.html -->
4  <!-- Linking to other web pages. -->
5  <html>
6  <head>
7      <meta charset = "utf-8">
8      <title>Links</title>
9  </head>
10
11 <body>
12     <h1>Here are my favorite sites:</h1>
13     <p><strong>Click a name to visit that site.</strong></p>
14
15     <!-- create four text hyperlinks -->
16     <p><a href = "http://www.facebook.com">Facebook</a></p>
17     <p><a href = "http://www.twitter.com">Twitter</a></p>
18     <p><a href = "http://www.foursquare.com">Foursquare</a></p>
19     <p><a href = "http://www.google.com">Google</a></p>
20 </body>
21 </html>
```





## 2.6 Linking (Cont.)

(語法很多，課堂講常用的)

**<strong> ... </strong>** 強調

- ▶ indicates the **content** with **high importance** (用於強調).
- ▶ Browsers render such texts in a **bold font** (以粗體顯示).

雙引號  
`<a href = "http://www.facebook.com">Facebook</a>`

- ▶ 語法範例: `<a href="連結網址">顯示文字</a>`
  - ▶ **a**: anchor(超連結起點).
  - ▶ **href**: hypertext reference(超連結路徑)，如：
    - ▶ a web page (外部連結) or location within a web page (內部連結)
    - ▶ a file (檔案下載)
    - ▶ an e-mail address (外部程式)
- ▶ Note: if a **URL** (Ex: `http://www.facebook.com/`) **does not** indicate a html5 document, a default web page **"index.html"** will be returned.
  - ▶ 註：未指定網頁時，將以預設**index.html**回傳。

## 2.6 Linking (Cont.)

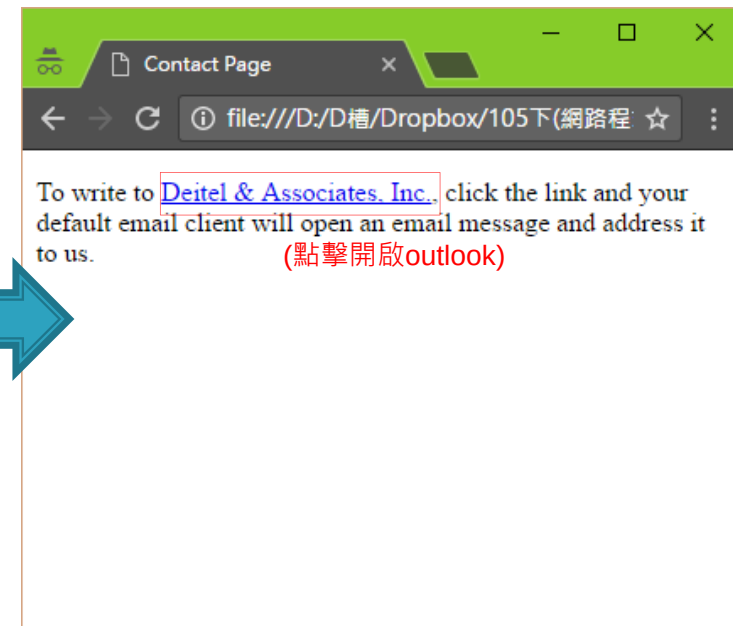
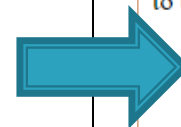
### *Hyperlinking to an E-Mail Address (超連結-email)*

- links to an e-mail address by mailto: URL (範例: "mailto: test@mail.nutn.edu.tw")
  - When clicking, the browsers launch the default e-mail program (e.g., Microsoft Outlook or Apple Mail) to write an e-mail message to the linked address.

```
4) contact.html x
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.4: contact.html -->
4  <!-- Linking to an e-mail address. -->
5  <html>
6  <head>
7      <meta charset = "utf-8">
8      <title>Contact Page</title>
9  </head>
10
11 <body>
12 <p>
13     To write to <a href = "mailto:deitel@deitel.com">
14     Deitel & Associates, Inc.</a>, click the link and your default
15     email client will open an email message and address it to us.
16 </p>
17 </body>
18 </html>
```

(練習trace程式碼)

(連結email)





# In-Class Exercise #1

- ▶ Create an **HTML5 document** that contains links to your **favorite Web sites**.
- ▶ 課中練習題：請同學建立一個html5文件，連結到自已喜愛的網頁(3個)。

(課堂最後 會預留時間練習)

## 2.7 Images

`<img src=“圖片來源位置”>... </img>` 嵌入圖片

- ▶ The most popular image formats on web are PNG (Portable Network Graphics) and JPEG (Joint Photographic Experts Group).
- ▶ You can create images by
  - ▶ Adobe Photoshop Express ([www.photoshop.com](http://www.photoshop.com)),
  - ▶ G.I.M.P. ([www.gimp.org](http://www.gimp.org)), \* 使用圖片及素材時，請留意版權
  - ▶ Inkscape (<https://inkscape.org/>) etc.
- ▶ Some royalty-free images (線上免費影像素材圖庫):
  - 💡 <https://www.shutterstock.com/> (舉例：cartoon)
  - 💡 <https://www.graphicstock.com/>
  - 💡 <https://unsplash.com/>

★(補充) 連結：[2025 設計師愛用的五個免費圖庫 + 設計工具](#)

## 2.7 Images

```
5) image.html x
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.6: picture.html -->
4  <!-- Including images in HTML5 files. -->
5  <html>
6  <head>
7      <meta charset = "utf-8">
8      <title>Images</title>
9  </head>
10
11 <body>
12 <p>
13     <img src = "cpphttp.png" width = "92" height = "120"
14         alt = "C++ How to Program book cover">
15     <img src = "jhttp.png" width = "92" height = "120"
16         alt = "Java How to Program book cover" />
17 </p>
18 </body>
19 </html>
```



```
<img src = "cpphttp.png" width = "92" height = "120"
    alt = "C++ How to Program book cover">
```

- ▶ **src="..."**: specifies an image's location (影像位置, 若未指定, 則同目前html同目錄)
- ▶ **width** and **height**: control image's width and height (非必填的屬性, 可控制縮小/放大, 單位: pixel 像素量)
- ▶ **alt="..."**: the displayed texts if the browser cannot display the image.

註: 當無法讀取圖片(或圖片損壞)時, 仍可知道其表達意涵

## 2.7 Images (Cont.)



### *<.../> Void Elements (單標籤)*

- ▶ The elements is not placed between a **start** and an **end tag**.
  - ▶ 註：當element不含外顯文字時，只有 **單標籤**
  - ▶ 使用方式：在 **>** 結束前，加上 斜線 **/**.

```
<img src = "jhttp.png" width = "92" height = "120"  
alt = "Java How to Program book cover" />
```

### *Using Images as Hyperlinks (超連結-圖片)*

- ▶ You can create **graphical link** (圖片超連結) to other resources.



## 2.7 Images (Cont.)

### ▶ Example 2:

```
5-1)nav.html x
1 <!DOCTYPE html>
2 D:\D槽\Dropbox\105下(網路程式設計)\Slides_編輯by0212\Part1\Ch02課程
3 <!-- Fig. 2.7: nav.html -->
4 <!-- Images as link anchors. -->
5 <html>
6 <head>
7 <meta charset = "utf-8">
8 <title>Navigation Bar</title>
9 </head>
10
11 <body>
12 <p>
13 <a href = "links.html">
14 <img src = "buttons/links.jpg" width = "65"
15 height = "50" alt = "Links">
16 </a>
17
18 <a href = "list.html">
19 <img src = "buttons/list.jpg" width = "65"
20 height = "50" alt = "List of Features">
21 </a>
22
23 <a href = "contact.html">
24 <img src = "buttons/contact.jpg" width = "65"
25 height = "50" alt = "Contact Me">
26 </a>
27
28 <a href = "table1.html">
29 <img src = "buttons/table.jpg" width = "65"
30 height = "50" alt = "Tables Page">
31 </a>
32
33 <a href = "form.html">
34 <img src = "buttons/form.jpg" width = "65"
35 height = "50" alt = "Feedback Form">
36 </a>
37 </p>
38 </body>
39 </html>
```

(5個影像超連結)



(課本範例)



(影像超連結)

## 2.8 Special Characters and Horizontal Rules



- ▶ **Special characters (特殊符號):**
  - for representing special characters in web pages
- ▶ **代碼規格:**
  - **Word abbreviations (縮寫) 或 Numbers (數字)**

| Symbol                          | Description  | Character entity reference |
|---------------------------------|--------------|----------------------------|
| HTML5 character entities        |              |                            |
| &                               | ampersand    | &amp;                      |
| '                               | apostrophe   | &apos;                     |
| >                               | greater-than | &gt;                       |
| <                               | less-than    | &lt;                       |
| "                               | quote        | &quot;                     |
| Other common character entities |              |                            |
| non-breaking space              |              | &nbsp; //空白                |
| ©                               | copyright    | &copy;                     |
| —                               | em dash      | &mdash;                    |
| –                               | en dash      | &ndash;                    |
| ¼                               | fraction 1/4 | &frac14;                   |
| ½                               | fraction 1/2 | &frac12;                   |

\* [補充] :

自W3School查詢特殊字元 :

[https://www.w3school.com.cn/tags/html\\_ref\\_entities.html](https://www.w3school.com.cn/tags/html_ref_entities.html)

**Fig. 2.8** | Some common HTML character entity references.



# [補充] : Special characters (特殊符號)

|   |           |   |          |   |           |   |         |   |           |
|---|-----------|---|----------|---|-----------|---|---------|---|-----------|
| ≈ | &asymp;   | / | &frasl;  | ← | &larr;    | ∂ | &part;  | ♠ | &spades;  |
| ∩ | &cap;     | ≥ | &ge;     | ≤ | &le;      | ” | &Prime; | Σ | &sum;     |
| ♣ | &clubs;   | ↔ | &harr;   | ◇ | &loz;     | ’ | &prime; | ↑ | &uarr;    |
| ↓ | &darr;    | ♥ | &hearts; | – | &minus;   | Π | &prod;  |   | &zwj;     |
| ◆ | &diams;   | ∞ | &infin;  | ≠ | &ne;      | √ | &radic; |   | &zwnj;    |
| ≡ | &equiv;   | ∫ | &int;    | ¯ | &oline;   | → | &rarr;  |   |           |
|   |           |   |          |   |           |   |         |   |           |
| α | &alpha;   | η | &eta;    | μ | &mu;      | π | &pi;    | θ | &theta;   |
| β | &beta;    | γ | &gamma;  | ν | &nu;      | ψ | &psi;   | υ | &upsilon; |
| χ | &chi;     | ι | &iota;   | ω | &omega;   | ρ | &rho;   | ξ | &xi;      |
| δ | &delta;   | κ | &kappa;  | ο | &omicron; | σ | &sigma; | ζ | &zeta;    |
| ε | &epsilon; | λ | &lambda; | φ | &phi;     | τ | &tau;   |   |           |
|   |           |   |          |   |           |   |         |   |           |
| A | &Alpha;   | H | &Eta;    | M | &Mu;      | Π | &Pi;    | Θ | &Theta;   |
| B | &Beta;    | Γ | &Gamma;  | N | &Nu;      | Ψ | &Psi;   | Υ | &Upsilon; |
| X | &Chi;     | I | &Iota;   | Ω | &Omega;   | P | &Rho;   | Ξ | &Xi;      |
| Δ | &Delta;   | K | &Kappa;  | O | &Omicron; | Σ | &Sigma; | Z | &Zeta;    |
| E | &Epsilon; | Λ | &Lambda; | Φ | &Phi;     | T | &Tau;   | ς | &sigmaf;  |

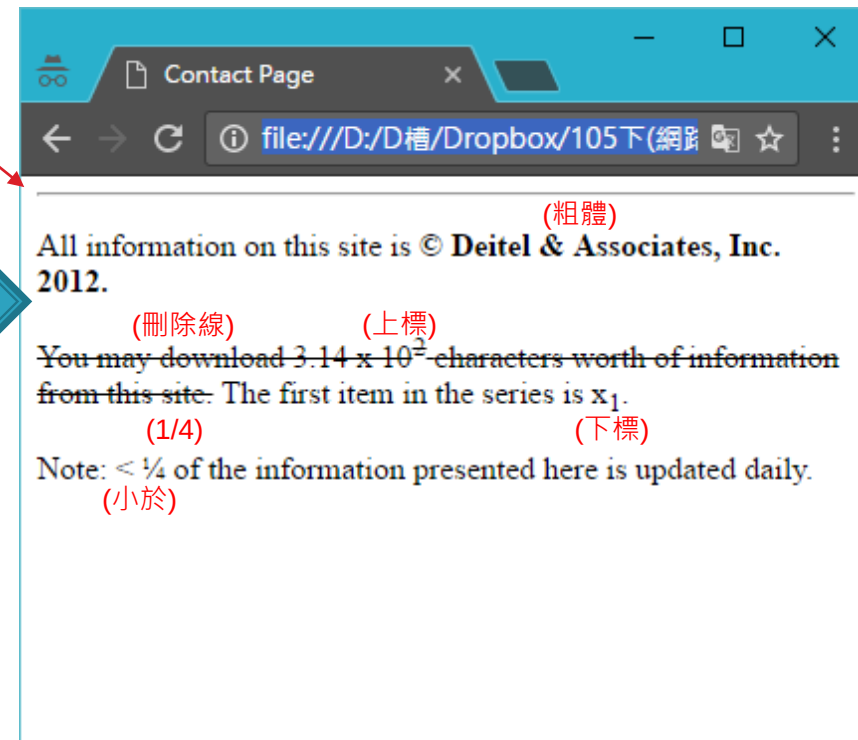
\* [補充] : 自W3School查詢特殊字元 :

[https://www.w3school.com.cn/charsets/ref\\_utf\\_symbols.asp](https://www.w3school.com.cn/charsets/ref_utf_symbols.asp)

## 2.8 Special Characters and Horizontal Rules

- ▶ **Special characters** can also be represented as **decimal** (10進位數值) or **hexadecimal values** (16進位數值).
  - Ex: “&” 也可表示成 &#38 (10進位) 或 &#x26 (16進位)
- 💡 參考網址：<https://www.ifreesite.com/html-entities.htm>
- horizontal rule: <hr>** 表示水平線 (早期用法).

```
6) code.html x
1 <!DOCTYPE html>
2
3 <html>
4   <head>
5     <meta charset = "utf-8">
6     <title>Contact Page</title>
7   </head>
8
9   <body>
10    <hr> <!-- inserts a horizontal rule -->
11
12    <p>All information on this site is <strong>&copy;
13      Deitel & Associates, Inc. 2012.</strong> </p>
14
15    <p><del>You may download 3.14 x 10<sup>2</sup>
16      characters worth of information from this site.</del>
17      The first item in the series is x<sub>1</sub>.</p>
18
19    <p>Note: <math>\frac{1}{4}</math> of the information (下標)
20      presented here is updated daily.</p>
21  </body>
22 </html>
```



## 2.9 Lists (序列)

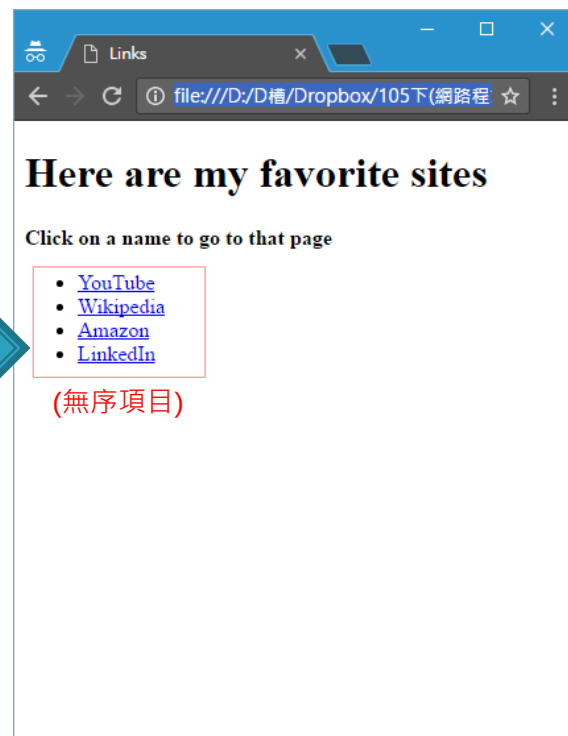
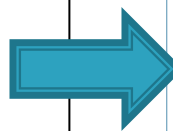


### `<ul>...</ul>` Unordered list element 無序項目

- 用途：產生一列表，但不含編號
- 使用方式：以`<li>...</li>` 包覆項目內容
  - 通常以“bullet (黑點)”作為開頭。

(舉例：新聞條列)

```
1 <!DOCTYPE html>
2
3 <!-- Fig. 2.10: links2.html -->
4 <!-- Unordered list containing hyperlinks. -->
5 <html>
6 <head>
7   <meta charset = "utf-8">
8   <title>Links</title>
9 </head>
10
11 <body>
12   <h1>Here are my favorite sites</h1>
13   <p><strong>Click on a name to go to that page</strong></p>
14
15   <!-- create an unordered list -->
16   <ul> (無序項目)
17     <!-- the list contains four list items -->
18     <li><a href = "http://www.youtube.com">YouTube</a></li>
19     <li><a href = "http://www.wikipedia.org">Wikipedia</a></li>
20     <li><a href = "http://www.amazon.com">Amazon</a></li>
21     <li><a href = "http://www.linkedin.com">LinkedIn</a></li>
22   </ul>
23 </body>
24 </html>
```



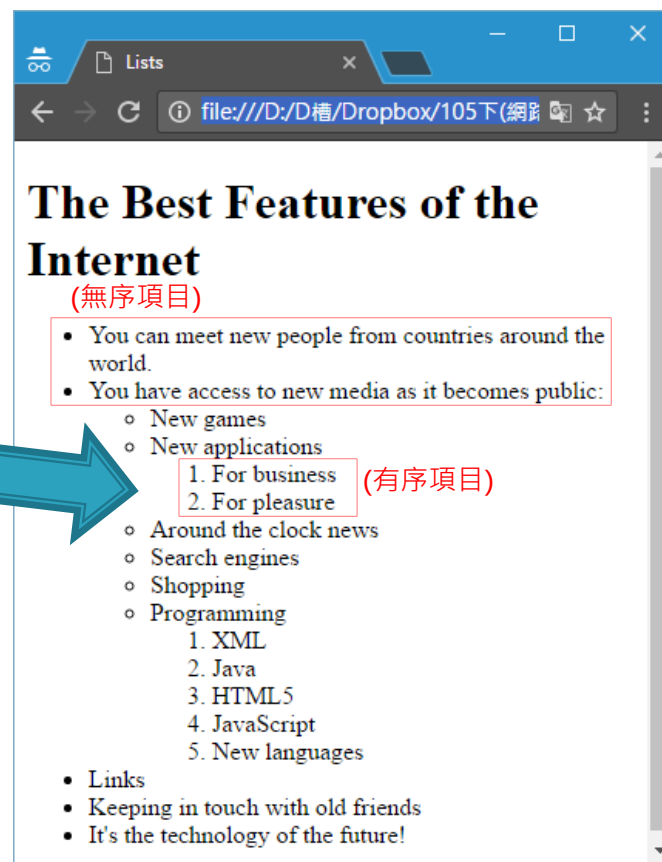
## 2.9 Lists (Cont.)



### *Nested Lists* (巢狀項目)

- ▶ To represent *hierarchical* (階層式) relationships.
- The ordered-list element `<ol>...</ol>` 有編號項目
  - 項目具有編號

```
14 <!-- create an unordered list --> (無序項目)
15 <ul>
16 <li>You can meet new people from countries around
17 the world.</li>
18 <li>
19 You have access to new media as it becomes public:
20
21 <!-- this starts a nested unordered list, which uses
22 <!-- different bullet. The list ends when you --> (無序項目)
23 <!-- close the <ul> tag. -->
24 <ul>
25 <li>New games</li>
26 <li>New applications
27
28 <!-- nested ordered list --> (有序項目)
29 <ol>
30 <li>For business</li>
31 <li>For pleasure</li>
32 </ol>
33 </li> <!-- ends line 27 new applica
34
35 <li>Around the clock news</li>
36 <li>Search engines</li>
37 <li>Shopping</li>
38 <li>Programming
39
40 <!-- another nested ordered list -->
41 <ol>
42 <li>XML</li>
43 <li>Java</li>
44 <li>HTML5</li>
45 <li>JavaScript</li>
46 <li>New languages</li>
47 </ol>
48 </li> <!-- ends programming li of line 38 -->
49 </ul> <!-- ends the nested list of line 24 -->
50 </li>
```



# 2.10 Tables 表格

## ▶ `<table> ... </table>` 表格

- used to **organize data** (組織性呈現資料) into *rows* and *columns*.

表格元素 : ■ `<caption>...</caption>` 表格標題 : 描述表格用途

### ■ Three sections (Table分三種區塊):

- Head (`<thead>...</thead>`) : 存放Table titles
- Body (`<tbody>...</tbody>`) : 存放Table data
- Table Foot (`<tfoot>...</tfoot>`) : 例如 : Calculation results (計算結果)、Footnotes (註腳)、注意事項
- `<tr>...</tr>` : "列(**raw**)"的內容
  - `<th>...</th>` : 標題或註腳的"內容欄"
  - `<td>...</td>` : 存放data的"內容欄"

(直接看後頁範例)

A screenshot of a web browser window titled "A simple HTML5 table". The browser shows a file:// URL. The table content is as follows:

| Fruit     | Price  |
|-----------|--------|
| Apple     | \$0.25 |
| Orange    | \$0.50 |
| Banana    | \$1.00 |
| Pineapple | \$2.00 |
| Total     | \$3.75 |

Labels with leader lines pointing to the table parts:

- Table caption: points to the text above the table.
- Table header: points to the first row of the table (th).
- Table body: points to the rows of data (tbody).
- Table footer: points to the last row of the table (tfoot).

```

11 <body>
12 <!-- the <table> tag opens a table -->
13 <table border = "1">
14
15 <!-- the <caption> tag summarizes the table's -->
16 <!-- contents (this helps visually impaired people) -->
17 <caption><strong>Table of Fruits (1st column) and
18   Their Prices (2nd column)</strong></caption> (caption)
19
20 <!-- the <thead> section appears first in the table -->
21 <!-- it formats the table header area -->
22 <thead> (head)
23   <tr> <!-- <tr> inserts a table row -->
24     <th>Fruit</th> <!-- insert a heading cell -->
25     <th>Price</th>
26   </tr>
27 </thead>
28
29 <!-- the <tfoot> section appears last in the table -->
30 <!-- it formats the table footer -->
31 <tfoot> (foot)
32   <tr>
33     <th>Total</th>
34     <th>$3.75</th>
35   </tr>
36 </tfoot>
37
38 <!-- all table content is enclosed -->
39 <!-- within the <tbody> -->
40 <tbody> (body)
41   <tr>
42     <td>Apple</td> <!-- insert a data cell -->
43     <td>$0.25</td>
44   </tr> (4列)
45   <tr>
46     <td>Orange</td> (一列中，有兩欄)
47     <td>$0.50</td>
48   </tr>
49   <tr>
50     <td>Banana</td>
51     <td>$1.00</td>
52   </tr>
53   <tr>
54     <td>Pineapple</td>
55     <td>$2.00</td>
56   </tr>
57 </tbody>
58 </table>
59 </body>

```

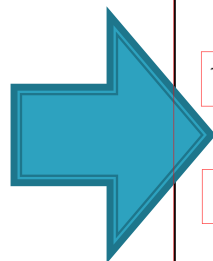


Table caption

Table header

Table body

Table footer

A simple HTML5 table

file:///

Links Other bookmarks Sync Error

| Table of Fruits (1st column) and Their Prices (2nd column) |        |
|------------------------------------------------------------|--------|
| Fruit                                                      | Price  |
| Apple                                                      | \$0.25 |
| Orange                                                     | \$0.50 |
| Banana                                                     | \$1.00 |
| Pineapple                                                  | \$2.00 |
| Total                                                      | \$3.75 |

– Table border

\* 容易混淆：

<thead>、<tfoot>的“欄位”：使用<th>  
<tbody>的“欄位”：使用<td>

## 2.10 Tables (Cont.)

(跨列、跨欄)

***rowspan="..." and colspan="..."*** (合併列、合併行 屬性)

- ▶ To merge data cells in row and column (合併多欄資料)
  - To specify the number of rows or columns occupied by the cell.

註：描述要併幾列、併幾行

- ▶ **<br>**：換行 (屬於單標籤)

```
8-2) table2.html
11 <body>
12 <h1>Table Example: Spanning Rows and Columns</h1>
13
14 <table border = "1">
15   <caption>A more complex sample table</caption>
16
17   <thead>
18     <!-- rowspans and colspans merge the specified -->
19     <!-- number of cells vertically or horizontally -->
20
21     <!-- merge two rows --> (合併2列)
22     <th rowspan = "2">
23       <img src = "camel.png" width = "205"
24       height = "167" alt = "Picture of a one-hump camel">
25     </th>
26
27     <!-- merge four columns --> (合併4行)
28     <th colspan = "4">
29       <strong>Camelid comparison</strong><br>
30       Approximate as of 6/2011
31     </th>
32
33   </thead>
34   <tbody>
35     <tr>
36       <th># of humps</th>
37       <th>Indigenous region</th>
38       <th>Spits?</th>
39       <th>Produces wool?</th>
40     </tr>
41
42     <tr>
43       <th>Camels (bactrian)</th>
44       <td>2</td>
45       <td>Africa/Asia</td>
46       <td>Yes</td>
47       <td>Yes</td>
48     </tr>
49
50     <tr>
51       <th>Llamas</th>
52       <td>1</td>
53       <td>Andes Mountains</td>
54       <td>Yes</td>
55       <td>Yes</td>
56     </tr>
57   </tbody>
58 </table>
```

(自行試試看!)

Tables

file:///D:/D槽/Dropbox/105下(網路程式設計)/Slides

### Table Example: Spanning Rows and Columns

A more complex sample table

|                                                                                              |                                                          |                   |        |                |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------|-------------------|--------|----------------|
| <br>(跨列) | Camelid comparison<br>Approximate as of 6/2011<br>(合併4欄) |                   |        |                |
|                                                                                              | # of humps                                               | Indigenous region | Spits? | Produces wool? |
| Camels (bactrian)                                                                            | 2                                                        | Africa/Asia       | Yes    | Yes            |
| Llamas                                                                                       | 1                                                        | Andes Mountains   | Yes    | Yes            |

# In-Class Exercise #2

- ▶ Write a HTML5 document that produces the following table.
- ▶ 課中練習題：請同學試著建立出以下Table。

| Student ID | Homework |    | Exam    |       | Grade |
|------------|----------|----|---------|-------|-------|
|            | #1       | #2 | Midterm | Final |       |
| B12345678  | 80       | 75 | 90      | 88    | 84    |
| B56789012  | 95       | 90 | 60      | 80    | 78    |

(想想看!)

## 2.11 Forms (表單)

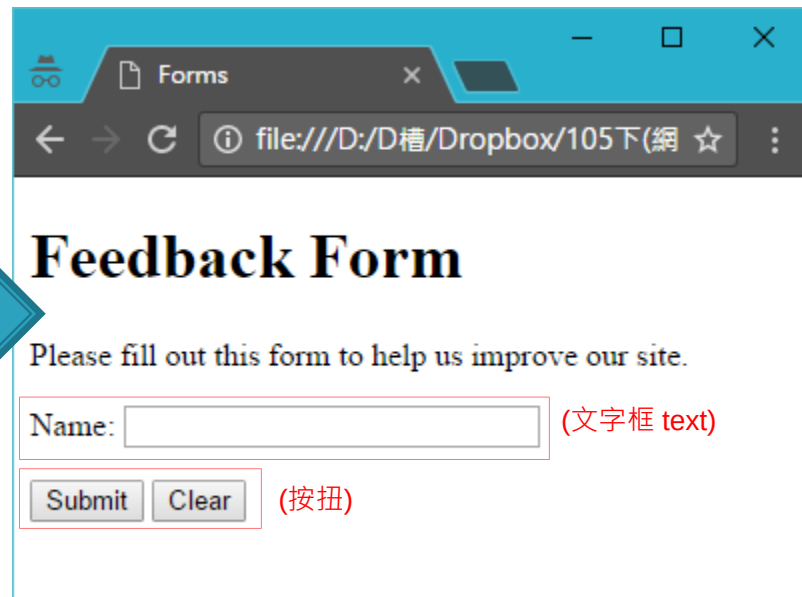
(用途：收集資訊)

### ▶ **<form> ... </form>** (表單)

- To collecting information from users.

• 應用範例：Google 搜尋、客戶意見回覆、購物規格...收集等

```
9) form.html x
16 (傳遞方法: get 或 post) (傳去哪)
17 <form method = "post" action = "http://www.deitel.com">
18 <!-- hidden inputs contain non-visual -->
19 <input type = "hidden" name = "recipient"
20 value = "deitel@deitel.com"> (隱藏-不顯示變數)
21 <input type = "hidden" name = "subject"
22 value = "Feedback Form">
23 <input type = "hidden" name = "redirect"
24 value = "main.html">
25
26 <!-- <input type = "text"> inserts a text field -->
27 <p><label>Name: (文字框)
28 <input name = "name" type = "text" size = "25"
29 maxlength = "30">
30 </label></p>
31
32 <p>
33 <!-- input types "submit" and "reset" insert -->
34 <!-- buttons for submitting and clearing the -->
35 <!-- form's contents, respectively -->
36 <input type = "submit" value = "Submit"> (按鈕)
37 <input type = "reset" value = "Clear">
38 </p>
39 </form>
```



Feedback Form

Please fill out this form to help us improve our site.

Name:



## 2.11 Forms (Cont.)

### *method="..." 回傳表單方式*

- To specify which “method” (上次提過，有“post”和“get”等2種傳遞方式) the data is sent to the web server.
- `method="post"` to send data with the protocol (HTTP) message.
  - 註：“post”是將內容透過http協定傳遞。
- `method="get"` to send data with the end of the URL, where it's visible in the browser's Address field.
  - 註：“get”是將傳遞內容夾帶在URL傳遞。
- `action="..."` 表單要傳送至哪個server
  - 註：之後server side-script 會詳細講
- `<input type="...">` 描述輸入型態，又稱“form handler”.

## 2.11 Forms (Cont.)

***<input type="hidden" ...>*** (隱藏-不顯示輸入)

- ▶ Such as **clickable buttons** (按鈕) and **some special data** (某些屬性，不特別顯示，ex: e-mail addresses).

***<input type="text" size="..." ...>*** (文字框)

- ▶ A **text field** allows the user to input data.
  - ▶ Ex: 回饋的意見內容(100字)。
  - ▶ **size="..."** specifies the **number of characters visible** in the text field.
    - ▶ 註：外觀可看見多少字 
  - ▶ **maxlength="..."** limits the **number of characters input** into a text field.
    - ▶ 註：實際可輸入多少字

(可修改按鈕顯示名稱，例如：value="送出")

***<input type="submit" value = "Submit">***：送出表單按鈕

***<input type="reset" value = "Clear">***：清除表單按鈕

- ▶ **value = "..."**：按鈕上顯示之文字

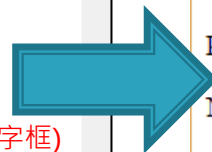
## 2.11 Forms (Cont.)



- ▶ **<textarea rows="..." cols="..." > ... </textarea>** 多行文字框
  - ▶ **rows="..."**: specified the number of rows (列數)
  - ▶ **cols="..."**: specified the number of columns (行數)
- ▶ **<input type="password" size="...">** 輸入隱私資訊(顯示星號)
  - Allows users to enter sensitive information (隱私、敏感資料), such as credit card numbers (信用卡號碼) and passwords (密碼)
    - By "masking" the information input, ex: **\*\*\*\*\*** (顯示星號).

\* 補充：可能透過抓封包而洩露 (因此現今多用https加密)

```
9-2) form2.html 9-1) form.html
16 <form method = "post" action = "http://www.deitel.com">
17
18 <input type = "hidden" name = "recipient"
19     value = "deitel@deitel.com">
20 <input type = "hidden" name = "subject"
21     value = "Feedback Form">
22 <input type = "hidden" name = "redirect"
23     value = "main.html">
24
25 <p><label>Name:
26     <input name = "name" type = "text" size = "25">
27 </label></p>
28
29 <!-- <textarea> creates a multiline textbox -->
30 <p><label>Comments:<br>
31     <textarea name = "comments"
32         rows = "4" cols = "36">Enter comments here.</textarea>
33 </label></p>
34
35 <!-- <input type = "password"> inserts a -->
36 <!-- textbox whose display is masked with -->
37 <!-- asterisk characters -->
38 <p><label>E-mail Address:
39     <input name = "email" type = "password" size = "25">
40 </label></p>
```



\* 補充：<label>...</label> 讓包覆內容具有相同性質 (ex: 點擊"包覆文字"等同點擊輸入"勾選")

# 2.11 Forms (Cont.)

(ftp可下載範例)

**<input type="checkbox">** 方框選項 (單選/複選)

▶ For users to **select** some **options** (提供使用者選取/選項).

- Can be used **individually** (單選) and in **groups** (複選).

- 註：複選時，需設定成相同**name**屬性
- Ex: 購買哪些書籍：C++, HTML, MySQL 等。

**<input type="radio">** 圓框選項 (單選)

- only one radio button** can be selected at any time (只能單選).

- All radio buttons in a group have the same name attribute but different value attributes.

- 註：可以設很多選項，但最後只能擇一選擇
- Ex: 買衣服尺寸：S, M, L, XL, ... 等。

**<select name = "rating">** 下拉式選單 (單選)

- Provides a **drop-down** (下拉式) list of items.
- <option>...</option>** 描述選項

Things you liked: (方框選項)  
Site design ☐ Links ☐ Ease of use ☐ Images ☐  
Source code ☐

How did you get to our site?: (圓框選項)  
Search engine ☒ Links from another site ☐  
Deitel.com Web site ☐ Reference in a book ☐ Other ☐

Rate our site: Amazing (下拉式選單-較精簡)

Submit Clear

```
9-2) form2.html
49      <input name = "thingsliked" type = "checkbox"
50      value = "Links"></label>
51      <label>Ease of use
54      <label>Images
57      <label>Source code
58      <input name = "thingsliked" type = "checkbox"
59      value = "Code"></label>
60
61      </p>
62      <p>
63      <strong>How did you get to our site?:</strong><br>
64
65      <label>Search engine
66      <input name = "howtosite" type = "radio"
67      value = "search engine" checked></label>
68      <label>Links from another site
69      <input name = "howtosite" type = "radio"
70      value = "link"></label>
71      <label>Deitel.com Web site
73      <label>Reference in a book
76      <label>Other
79
80      </p>
81
82      <p>
83      <label>Rate our site:
84      <select name = "rating">
85      <option selected>Amazing</option>
86      <option>10</option>
87      <option>9</option>
88      <option>8</option>
89      <option>7</option>
90      <option>6</option>
91      <option>5</option>
92      <option>4</option>
93      <option>3</option>
94      <option>2</option>
95      <option>1</option>
96      <option>Awful</option>
97      </select>
98      </label>
99
100     <p>
101     <input type = "submit" value = "Submit">
102     <input type = "reset" value = "Clear">
103     </p>
```

\* **補充**: <label>...</label> 讓包覆內容具有相同性質  
(ex: 點擊"包覆文字"等同點擊輸入"勾選")



# In-Class Exercise #3

- ▶ Write an HTML5 document that produces the following web page.
- ▶ 課中練習題：請同學試著建立出以下表單

Chapter 2: In-Class Exercise / ✕

file:///Z:/CGU/Courses/101下\_網際網路程式設計/www/slides/

訂閱... Google Bookmarks 其他書籤

### 教科書預訂系統

姓名:

寢室分機:

選擇書籍:

- ☐ Internet & World Wide Web: How to Program
- ☐ Visual C# 2010: How to Program
- ☐ Data Structure

付款方式:

- ☒ Cash
- ☐ Check
- 信用卡:
  - ☐ Mastercard
  - ☐ Visa
  - ☐ American Express

確認訂單，請按下此按鈕

PS. 請指定-測試server位址(如下)，以驗證表單輸入結果:  
`action="https://120.114.190.22/web_course/get_post.php"`

# 2.12 Internal Linking

## ▶ 內部連結 (Hyper Link) :

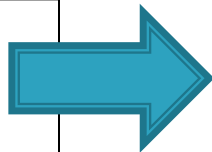
`<h1 id="名稱">...</h1>` 定義內頁連結id名稱

- To link to another section (連到本頁另一部分) of the same document by specifying the element's id as the link's href. 例如：網頁文章很長、有很多章節，可使用

`<a href = "#名稱">` 連結到標示" id=名稱"的位置

- To link internally to an element with its id attribute set, use the syntax `#id`.
  - 註：透過定義id位置，便可使用超連結連至該位置。

```
11 <body>
12 <!-- id attribute creates an internal hyperlink destination -->
13 <h1 id = "features">The Best Features of the Internet</h1>
14
15 <!-- an internal link's address is "#id" -->
16 <p><a href = "#bugs">Go to <em>Favorite Bugs</em></a></p>
17
18 <ul>
19 <li>You can meet people from countries
20 around the world.</li>
21 <li>You have access to new media as it becomes public:
22
23 <li>Links</li>
24 <li>Keeping in touch with old friends</li>
25 <li>It is the technology of the future!</li>
26 </ul>
27
28 <!-- id attribute creates an internal hyperlink destination -->
29 <h1 id = "bugs">My 3 Favorite Bugs</h1>
30 <p>
31 <!-- internal hyperlink to features -->
32 <a href = "#features">Go to <em>Favorite Features</em></a>
33 </p>
34 <ol>
35 <li>Fire Fly</li>
36 <li>Gal Ant</li>
37 <li>Roman Tic</li>
38 </ol>
39 </body>
```



## 2.13 meta Elements



**<meta name="..." content="...">** 提供搜尋引擎、查詢分類

- ▶ For search engines catalog pages is by reading the meta element's contents.
  - **name="keywords" Content="..."**: provides **keywords** for search engines with a **list of words** that describe a page.
    - 註：提供給搜尋引擎一些關鍵字，描述此頁網站資訊
  - **name="description" Content="..."**: provides **descriptions** for **search engines** to catalog your site. This text is sometimes **displayed** as the **search result**
    - 註：提供給搜尋引擎一些描述，以作為網站分類 (也作為搜尋結果顯示)

```
11) meta.html
4  <!-- meta elements provide keywords and a description of a page. -->
5  <html>
6  <head>
7    <meta charset = "utf-8">
8    <title>Welcome</title>
9
10   <!-- <meta> tags provide search engines with -->
11   <!-- information used to catalog a site -->
12   <meta name = "keywords" content = "web page, design,
13     HTML5, tutorial, personal, help, index, form,
14     contact, feedback, list, links, deitel">
15   <meta name = "description" content = "This website will
16     help you learn the basics of HTML5 and web page design
17     through the use of interactive examples and
18     instruction.">
19 </head>
20 <body>
21   <h1>Welcome to Our Website!</h1>
22
23   <p>We have designed this site to teach about the wonders
24     of <strong><em>HTML5</em></strong>. <em>HTML5</em> is
25     better equipped than <em>HTML</em> to represent complex
26     data on the Internet. <em>HTML5</em> takes advantage of
27     XML's strict syntax to ensure well-formedness. Soon you
28     will know about many of the great features of
29     <em>HTML5.</em></p>
30
31   <p>Have Fun With the Site!</p>
32 </body>
33 </html>
```

(提供網站分類)

(描述網頁資訊)





# Lab

## (上機練習)

# In-Class Exercise #1

- ▶ Create an **HTML5 document** that contains links to your **favorite Web sites**.
- ▶ 課中練習：請同學建立一個html5文件，連結到自已喜愛的網頁(3個)。

- 舉例如下：



# In-Class Exercise #2

- ▶ Write a HTML5 document that produces the following table.
- ▶ 課中練習：請同學試著建立出以下Table。

(需要熟悉一下)

//提示: caption

Chun-Liang

Midterm Q1

file:///D:/Dropbox/Courses/103下\_網際網路程式設計/www

**Grade Sheet**

| Student ID | Homework |    | Exam    |       | Grade |
|------------|----------|----|---------|-------|-------|
|            | #1       | #2 | Midterm | Final |       |
| B12345678  | 80       | 75 | 90      | 88    | 84    |
| B56789012  | 95       | 90 | 60      | 80    | 78    |



# In-Class Exercise #3

- ▶ Write an HTML5 document that produces the following web page.
- ▶ 課中練習：請同學試著建立出以下訂書表單

Chapter 2: In-Class Exercise / x

file:///Z:/CGU/Courses/101下\_網際網路程式設計/www/slides/

訂閱... Google Bookmarks 其他書籤

### 教科書預訂系統

姓名：

寢室分機：

選擇書籍：

- ☐ Internet & World Wide Web: How to Program
- ☐ Visual C# 2010: How to Program
- ☐ Data Structure

付款方式：

- ☒ Cash
- ☐ Check
- 信用卡：
  - ☐ Mastercard
  - ☐ Visa
  - ☐ American Express

確認訂單，請按下此按鈕

- Ps. 請指定server端接收位址，驗證表單輸入結果：  
`action="https://120.114.190.22/web_course/get_post.php"`

ps. HTML5相關語法，鼓勵同學練習 / 上網查詢用法

- 注意：  
請於今日23:59 前上傳至ftp (否則算遲交)：  
<ftp://111webdesign:webdesign@120.114.190.22:4000>
- Lab02
- 請以“學號”建立資料匣 (否則扣分)



# \*課輔群組 / 補救教學

## (每週三13:00-15:00)

- 掃描QRcode，加入課輔群組~!

網頁程式設計 上課座位表 (簽名)



課輔群組(加入)

\_\_\_\_年 \_\_\_\_月 \_\_\_\_日

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